

HierMRL: Hierarchical Structure-Aware Molecular Representation Learning for Property Prediction

Maotao Liu¹, Yifan Yang², Xu Gong¹, Li Liu¹, Qun Liu^{1*}

¹ *Chongqing Key Laboratory of Computational Intelligence,
Chongqing University of Posts and Telecommunications, Chongqing China*
Email: {s200231202, d210201006}@stu.cqupt.edu.cn, {liliu, liuqun}@cqupt.edu.cn

² *Tencent, Shenzhen China*
Email: tobyfyang@tencent.com

Abstract—Deep graph neural networks have recently demonstrated powerful representation learning capabilities in bioinformatics. It is still crucial and challenging to design a representation model fusing the fundamental chemistry and biology knowledge. However, most existing representation models not only ignore domain knowledge but are also built on labeled datasets. To address this issue, this paper proposed a hierarchical structure-aware pre-training model that used contrastive learning to improve molecular representations with unlabeled datasets. We conducted comprehensive experiments on 13 molecular benchmark datasets from different application domains. The results demonstrate that our hierarchical structure-aware pre-trained model achieves superior performance against state-of-the-art baselines.

Index Terms—Molecule Modeling, Graph Neural Network, Molecular Representation Learning, Contrastive Learning

I. INTRODUCTION

Deep neural networks are used in bioinformatics to learn molecular representations [1]–[6] to avoid the high cost and time of traditional methods. It mainly includes two strategies. Firstly, molecules are used as text data, and representation learning is implemented by using models related to Natural Language Processing (NLP). Secondly, molecules are employed as graph data, and representation learning is carried out by using techniques such as Graph Neural Networks (GNNs).

Text-based methods exploit sequential representations of molecules, including Simplified Molecular Input Line Input System (SMILES) [7] and SELF-referencing Embedded Strings (SELFIES) [8]. SMILES-BERT [9] uses BERT [10] to do a pre-training task of predicting masked atoms. Chemformer [11] focuses on transformers constrained by molecular atomic connection information.

Graph-based methods [12]–[19] exploit the topology of molecules. Pre-GNN [16] pre-trained nodes and graphs using the graph neural network as an encoder. GeomGCL [20] utilized a multi-view framework to perform pre-training based on the correspondence and consistency of the molecule's 2D and 3D views. However, there is still potential for improvement

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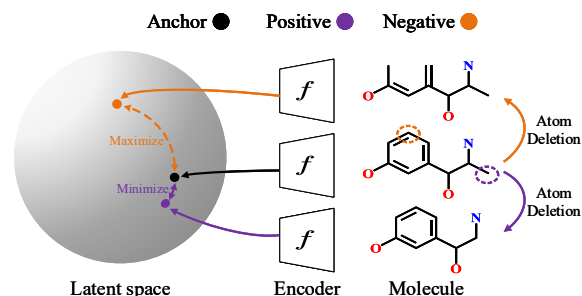


Fig. 1. For different types of augmented molecules (positive, negative), the model aims to maximize the distance between the anchor molecule and the negative molecule while minimizing the distance between the anchor molecule and the positive molecule.

in the current research on learning molecular representations, thus we propose a novel and improved method.

In this paper, we propose a novel contrastive learning-based method for improving the performance of molecular property prediction. Its primary contributions are outlined below:

- By using backbone and side-chain information, we present a hierarchical structure-aware molecular augmentation approach. An example is shown in Fig. 1.
- Based on augmented samples, we develop a pre-training framework HierMRL based on contrastive learning to obtain molecular representations.
- Extensive experiments on public datasets show that our pre-training framework achieves competitive results on several downstream tasks.

II. METHODS

This section describes our proposed hierarchical structure-aware representation learning model, consisting of three parts: preliminaries, problem definition, and framework analysis.

A. Preliminaries

1) *Molecular Graph*: Molecules can be regarded as graph data $G = (X, A)$ consisting of atoms (nodes) and chemical bonds (edges). $X \in \{0, 1\}^{n \times k}$ is the node feature matrix, and $A \in \{0, 1\}^{n \times n \times c}$ is the adjacency matrix. In which, n denotes the number of nodes of graph G , k and c represent the number of classes of nodes and edges, respectively.

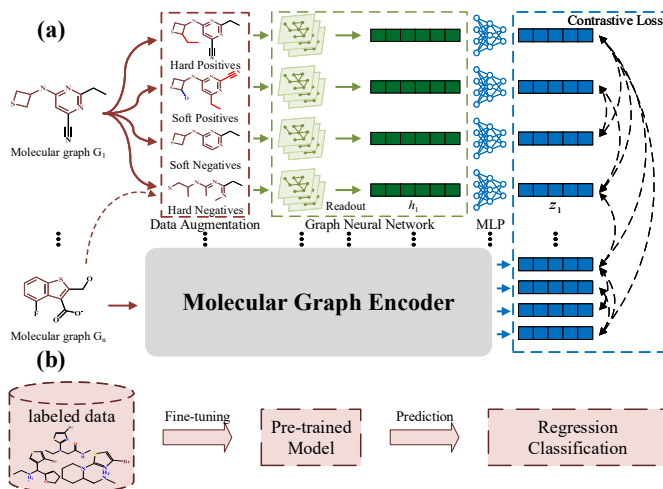


Fig. 2. Overview of HierMRL. (a) HierMRL pre-training: Reconstruct n molecule graphs as the anchors from the datasets as the batch. Four different data augment strategies are applied to each anchor molecular so as to obtain four different augmented samples. Each augmented sample and original sample are encoded using graph neural networks and pooling operations, then the final representation vectors of molecules are got using MLP with the contrastive loss. (b) HierMRL fine-tuning: Pre-trained models are fine-tuned for different downstream tasks using labeled data.

2) *Molecular Property Prediction*: Given a set of labeled graphs G^L with corresponding label y and a set of unlabeled graphs G^U , we aim to train a model to predict the labels of unlabeled graphs G^U .

B. Problem Definition

The objective of our model is to learn expressive GNNs by pre-training with unlabeled datasets, and the pre-trained GNN models can be fine-tuned to achieve competitive performance on a variety of downstream tasks.

C. Our Framework

As depicted in Fig. 2, HierMRL¹ consists of three components: graph neural network, data augmentation, and contrastive loss.

1) *Graph Neural Network*: Two different graph neural network models were used to learn the latent space representation of molecules, including Graph Convolutional Network (GCN) and Graph Isomorphism Network (GIN).

2) *Data Augmentation*: Data augmentation was developed to address the deficiency and imbalance of data. Currently, data augmentation can be used to obtain augmented samples for model training.

Molecular structures are naturally composed of two parts: the backbone and the side-chains. The backbone in the molecule determines its mass and geometry. Meanwhile, various side-chains correspond to various functional groups, which also influence the molecule's chemical properties. As shown in Fig. 3, four types of augmented samples can be obtained based on the perturbation of the backbone and side-chains.

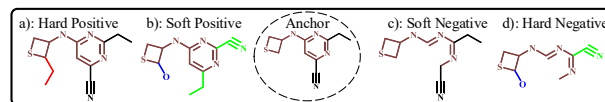


Fig. 3. Molecular Augmentation Strategies: side-chain repetition, side-chain generation, backbone disruption, and backbone disruption&side-chain deletion, are used to generate hard positive, soft positive, soft negative and hard negative samples, respectively.

3) *Contrastive Loss*: The goal of contrastive learning is to maximize the similarity of positive sample pairs while minimizing the similarity of negative sample pairs. The training objective can be formulated as follows:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z, z_+)/\tau)}{\exp(\text{sim}(z, z_+)/\tau) + \sum_{z_-} \exp(\text{sim}(z, z_-)/\tau)} \quad (1)$$

where z and z_+ are the anchor and its positive samples, z and z_- are the anchor and its negative samples, τ is the temperature parameter. For $\text{sim}(\cdot)$, we designed weighted cosine similarity to evaluate the similarity of sample pairs as follows:

$$\text{sim}(z_1, z_2) = e^{-d(z_1, z_2)} \cdot z_1^\top \cdot z_2 / (\|z_1\| \cdot \|z_2\|), \quad (2)$$

where $d(z_1, z_2)$ represents the Euclidean distance between the vectors z_1 and z_2 .

We denote the latent space vectors learnt from the anchor samples and four different types of augmented samples as z , z_{hp} , z_{sp} , z_{sn} , z_{hn} , representing the vectors of anchor, hard positive, soft positive, soft negative, and hard negative samples, respectively. Next, the similarity calculation is defined as follows:

$$\mathcal{S}_{p,q} = \sum_{q:p \in \mathcal{P}, q \in \mathcal{Q}} \exp(\text{sim}(p, q)/\tau) \quad (3)$$

$$\mathcal{R}_{p,q} = \sum_{r:p \in \mathcal{P}, r \in \mathcal{P} \cup \mathcal{Q}} \mathbb{I}(p \neq r) \exp(\text{sim}(p, r)/\tau) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function, $\mathcal{P}$ and $\mathcal{Q}$ are the set of p and q . Based on (3)-(4), the final contrastive loss function based on hard positive and anchor samples is formalized as:

$$\mathcal{L}_{z, z_{hp}} = -\log \frac{\exp(\text{sim}(z, z_{hp})/\tau)}{\mathcal{R}_{z, z_{hp}} + \alpha \mathcal{S}_{z, z_{sn}} + \beta \mathcal{S}_{z, z_{hn}}}, \quad (5)$$

where α , β are hyperparameters to control the weights of negative samples. The contrastive loss of soft positive samples and anchor samples and the contrastive loss of hard positive samples and soft positive samples can be calculated analogously to that of hard positive samples. In summary, the total loss function $\mathcal{L}$ of our model is as following:

$$\mathcal{L} = \mathcal{L}_{z, z_{hp}} + \lambda \mathcal{L}_{z, z_{sp}} + \mu \mathcal{L}_{z_{hp}, z_{sp}}, \quad (6)$$

where λ , μ are the hyperparameters used to balance several losses.

III. EXPERIMENTS AND DISCUSSION

In this section, the experimental setup and results are described.

¹<https://github.com/liumt021/HierMRL>.

TABLE I
THE PERFORMANCE OF THE DIFFERENT MODELS ON SEVEN CLASSIFICATION (ROC-AUC) AND SIX REGRESSION (RMSE & MAE) BENCHMARKS.

Dataset #Molecules #Tasks	Classification						
	BBBP 2039 1	Tox21 7831 12	ClinTox 1478 2	HIV 41127 1	BACE 1513 1	SIDER 1427 27	MUV 93087 17
GCN	71.8±0.9	70.9±2.6	62.5±2.8	74.0±3.0	71.6±2.0	53.6±3.2	71.6±4.0
GIN	65.8±4.5	74.0±0.8	58.0±4.4	75.3±1.9	70.1±5.4	57.3±1.6	71.8±2.5
SchNet	84.8±2.2	77.2±2.3	71.5±3.7	70.2±3.4	76.6±1.1	53.9±3.7	71.3±3.0
MGCN	85.0±6.4	70.7±1.6	63.4±4.2	73.8±1.6	73.4±3.0	55.2±1.8	70.2±3.4
D-MPNN	71.2±3.8	68.9±1.3	90.5±5.3	75.0±2.1	85.3±5.3	63.2±2.3	76.2±2.8
Pre-GNN	70.8±1.5	78.7±0.4	78.9±2.4	80.2±0.9	85.9±0.8	65.2±0.9	81.4±2.0
N-Gram	91.2±3.0	76.9±2.7	85.5±3.7	83.0±1.3	87.6±3.5	63.2±0.5	81.6±1.9
MolCLR _{GIN}	73.6±0.3	72.3±0.7	88.9±1.7	74.8±1.1	80.5±0.3	61.3±1.1	73.9±2.2
HierMRL _{GCN}	74.6±2.2	76.1±1.2	83.0±1.7	78.4±1.4	83.6±2.0	67.2±1.4	90.4±2.4
HierMRL _{GIN}	74.5±1.6	79.2±0.6	96.5±1.5	78.2±1.1	87.7±1.7	68.6±1.1	88.0±1.0

Dataset #Molecules #Tasks	Regression					
	FreeSolv (RMSE) 642 1	ESOL (RMSE) 1128 1	Lipo (RMSE) 4200 1	QM7 (MAE) 6830 1	QM8 (MAE) 21786 12	QM9 (MAE) 130829 8
GCN	2.87±0.14	1.43±0.05	0.85±0.08	122.9±2.2	0.0366±0.0011	5.796±1.969
GIN	2.76±0.18	1.45±0.02	0.85±0.07	124.8±0.7	0.0371±0.0009	4.741±0.912
SchNet	3.22±0.76	1.05±0.06	0.91±0.10	74.2±6.0	0.0204±0.0021	0.081±0.001
MGCN	3.35±0.01	1.27±0.15	1.11±0.04	77.6±4.7	0.0223±0.0021	0.050±0.002
D-MPNN	2.18±0.91	0.98±0.26	0.65±0.05	105.8±13.2	0.0143±0.0022	3.241±0.119
Pre-GNN	2.83±0.12	1.22±0.02	0.74±0.00	110.2±6.4	0.0191±0.0003	4.349±0.061
N-Gram	2.51±0.19	1.10±0.03	0.88±0.12	125.6±1.5	0.0320±0.0032	7.636±0.027
MolCLR _{GIN}	2.83±0.20	1.29±0.01	0.69±0.08	68.5±2.0	0.0183±0.0013	3.496±0.118
HierMRL _{GCN}	2.74±0.15	1.37±0.04	0.75±0.01	71.62±4.3	0.0182±0.0052	7.400±0.032
HierMRL _{GIN}	2.48±0.17	1.09±0.05	0.69±0.01	76.44±2.8	0.0178±0.0016	2.983±0.030

TABLE II
RESULTS OF HIERMRL ABLATION EXPERIMENTS ON CLASSIFIED BENCHMARKS. ✓ DENOTES DIFFERENT COMBINATIONS OF ENHANCEMENT STRATEGIES.

Augmentation Strategy			Tasks	
Hard Positives	Soft Positives	Negatives	Classification Avg.	Regression Avg.
-	✓	✓	75.01	2.050
✓	-	✓	75.99	1.777
✓	✓	-	77.43	1.666
✓	✓	✓	81.81	1.452

A. Datasets and Baselines

1) *Datasets*: For the model pre-training, we employ the 10 million unique unlabeled molecules obtained by ChemBERTa [21] from PubChem [22]. Then, RDKit [23] is used to build a molecular graph and extract chemical features from the SMILES string. To evaluate the performance of the HierMRL framework, thirteen MoleculeNet datasets [24] are used.

2) *Baselines*: i) Supervised learning methods: GCN [13], GIN [12], SchNet [25], D-MPNN [26] and MGCN [27]. ii) Self-supervised learning methods: N-Gram [28], Pre-GNN [16], and MolCLR [19].

B. Results and Analysis

In this section, we demonstrate the effectiveness of our model with the following experimental results.

1) *Molecular Property Predictions*: The molecular property prediction task is a well-established method of evaluating molecular representation models. We evaluated the model’s performance on 85 different classification and regression tasks in Table I. Classification tasks: Compared with the supervised model, HierMRL outperforms almost all supervised learning models. Compared to the self-supervised model, HierMRL achieves the best performance in five of the seven classification datasets. Regression tasks: Compared to the supervised

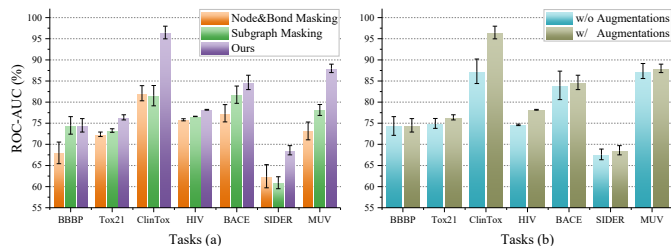


Fig. 4. Experimental results of an augmentation strategy for molecular classification benchmarks. w/o and w/ are abbr. for without and with, respectively.

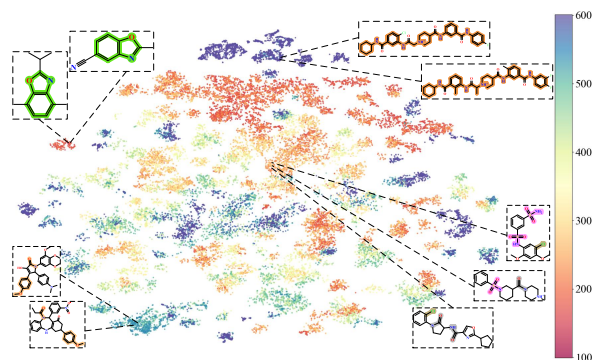


Fig. 5. Molecular representations learned by HierMRL are visualized after dimensionality reduction by t-SNE [29]. There are 50,000 molecules in this Figure. Data points are colored with their corresponding molecular weights.

learning model, HierMRL significantly outperforms the original GCN and GIN models. Compared to the self-supervised model, HierMRL achieves the best performance in five of the six datasets. Generally, the results show that by focusing on the hierarchical structure of the molecular graph, the model can capture the internal structure of the molecule more accurately, and result in higher performance.

2) *Analysis of Data Augmentation Strategies*: In this section, the data augmentation strategies proposed in this work are compared with traditional techniques. Meanwhile, we explore the effect of introducing augmented samples. Experiments were performed on seven classification datasets, using ROC-AUC values as metrics to measure experimental results, including means and standard deviations. The experimental results are depicted in Fig. 4. The results demonstrate that the augmentation strategy utilized in this study yields superior experimental outcomes and a smaller standard deviation. Simultaneously, this paper’s introduction of augmented samples can improve the experimental performance.

3) *Analysis of Ablation Experiments*: In this section, ablation experiments were used to evaluate the efficacy of each component. Table II shows the average results on classification and regression datasets (w/o QM7). From it, the results show that the augmented samples obtained by different strategies in HierMRL contribute to the model’s excellent performance.

4) *Visual Analysis of Molecular Representations*: We visualized the distribution of molecular representations in the latent space. Fig. 5 depicts the visualization results. According to the

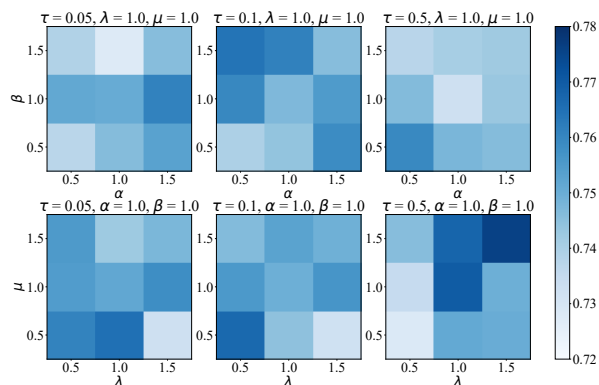


Fig. 6. AOC-AUC (%) metrics for different combinations of hyperparameters in the BBBP dataset.

diagram, molecules with similar weights are clustered together in the latent space. Meanwhile, the model can map molecules with similar backbone structures and functional groups closer in the latent space.

5) *Hyperparametric Search Experiments*: Fig. 6 depicts the experimental results of hyperparameters sensitivity. The experiments were performed on the BBBP dataset, and the ROC-AUC metric was utilized. The three subplots in the top row investigate the impact of the combination τ , α , β for our model. The three subplots in the second row investigate the influence of the combination of τ , λ and μ on the performance of the model. Therefore, we generalized the hyperparameter combinations $\tau = 0.1$, $\lambda = 1.5$, and $\mu = 1.5$, $\alpha = 0.5$, $\beta = 1.5$ to all datasets for the experiments.

IV. CONCLUSION

In this work, we propose HierMRL: a new molecular representation learning model based on hierarchical structure awareness for property prediction. Specifically, we explore four molecular graph augmentation strategies based on the hierarchical structure of molecules: side-chain repetition, side-chain addition, side-chain deletion, and backbone destruction. Moreover, A weighted contrastive loss is proposed for the augmented samples generated by different strategies, and a weighted cosine similarity formula is developed to accurately calculate the distance between augmented and anchor samples. Experiments demonstrate that the HierMRL pre-trained GNN model significantly outperforms various molecular benchmarks, which can greatly facilitate applications in fields such as drug discovery.

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